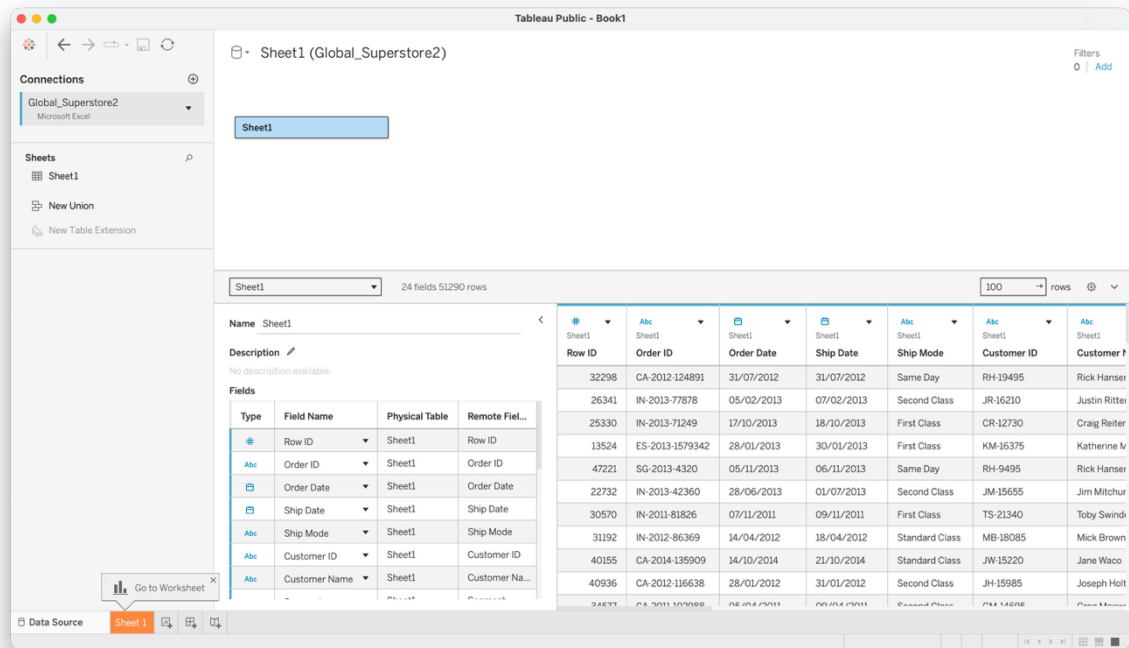


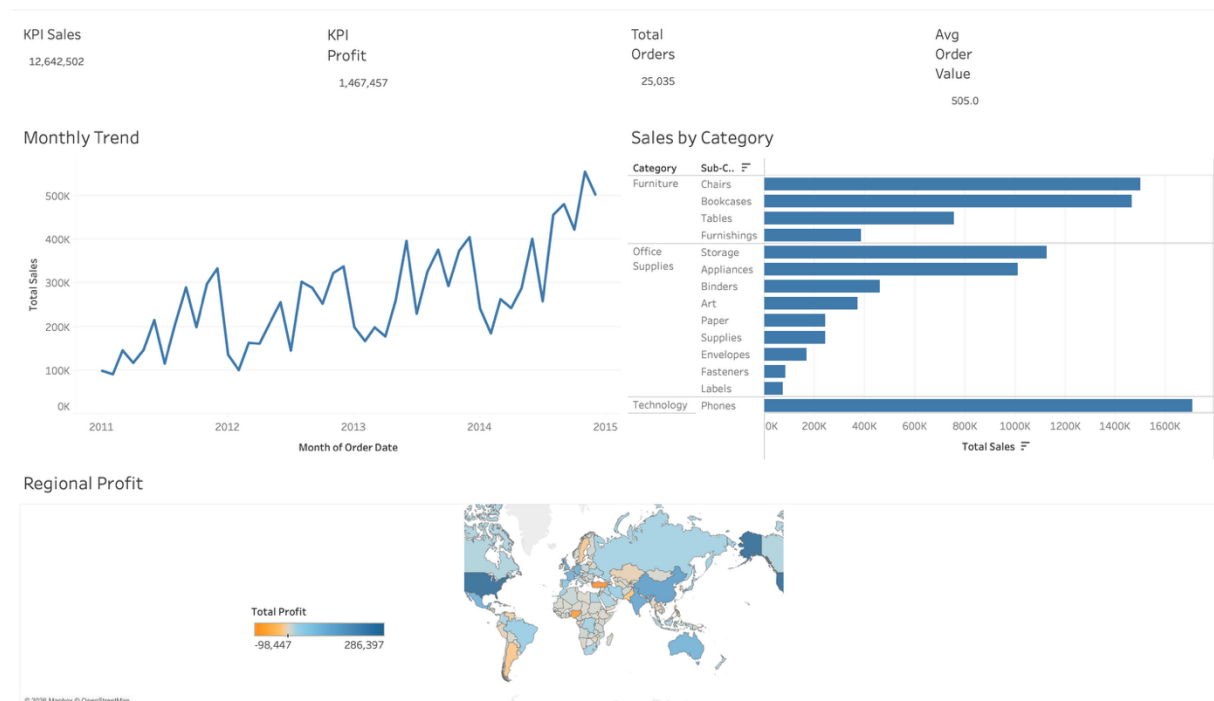
Lab Sheet – 05

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Section- B, Roll.No. 05

Dataset in Tableau:



Final Dashboard:



Questions and Answers

1. What is Business Intelligence (BI)? How does it support organizational decision-making?

- **Definition:** Business Intelligence (BI) refers to the strategies, architectures, and technology tools used to transform raw enterprise data into actionable visual insights.
- **Decision Support:** BI replaces guesswork with data-backed decisions by providing historical benchmarking, operational monitoring, predictive analytics, and real-time operational visibility across business units.

2. What is an interactive dashboard? How is it different from a static report?

- **Interactive Dashboard:** A dynamic visual interface where metrics, charts, and key performance indicators are interconnected, allowing users to apply slicers, drill down into hierarchical categories, and filter datasets on demand.
- **Static Report:** A fixed, point-in-time document (e.g., PDF or slide deck) displaying static summary numbers without interactive exploration capabilities.

3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

Feature / Aspect	Microsoft Power BI	Tableau Public / Desktop
Ecosystem & Integration	Native integration with Microsoft 365, Azure, and Excel	Strong integration with enterprise data warehouses and custom APIs
Data Transformation	Power Query (M code) and DAX for formula calculations	Tableau Prep Builder and VizQL calculation engine
Usability	Office-like ribbon interface; lower learning curve for Excel users	Intuitive drag-and-drop shelf interface with high custom visual control
Primary Focus	Cost-effective, enterprise-wide reporting and dashboarding	High-performance custom data exploration and advanced visualization

4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

- **Definition:** Quantifiable values used to evaluate how effectively a company or unit is achieving key business objectives.
- **Sales Analytics Examples:**
- **Total Revenue (Sales):** Net monetary value generated over a target period.
- **Customer Lifetime Value (CLV):** Total revenue expected from a single customer account.
- **Customer Acquisition Cost (CAC):** Total cost spent to acquire a new customer.
- **Profit Margin Ratio:** Percentage of sales revenue remaining after operational costs.

5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

- **Filters:** Restrict data at the visual, page, or dashboard level based on predefined backend logic.
- **Slicers:** On-canvas UI components (e.g., dropdowns, date sliders) enabling users to modify dashboard parameters interactively.
- **Drill-Down:** Allows users to navigate down through hierarchical data layers (e.g., Year → Quarter → Month, or Category → Sub-Category → Product).

6. What factors should be considered while designing an effective business dashboard?

- **Audience Orientation:** Design specifically for the end-user's operational needs (e.g., high-level KPIs for executives vs. detailed transactional drill-downs for managers).
- **Visual Hierarchy:** Position primary metrics (KPI cards) at the top left, followed by primary trends and secondary geographic or structural distributions.
- **Color Psychology:** Use purposeful color palettes consistently (e.g., avoiding red/green ambiguity for colorblind accessibility).
- **Clutter Reduction:** Avoid overcrowding canvas space with redundant visuals or complex chart types.

7. How can dashboards help managers monitor organizational performance in real time?

- Direct connections to live database servers (e.g., via Power BI Ans- DirectQuery or Tableau Live Connections) update dashboards automatically as new transactions occur.
- Automated alert mechanisms send notifications when performance metrics drop below predefined operational thresholds, enabling fast response times.

8. Why is data visualization important in Business Intelligence tools?

Ans- Data visualization processes dense tabular records into intuitive graphic patterns, making complex structural trends, anomalies, season variations, and correlations easy to understand for organizational decision-makers.

9. What insights can be derived from a sales dashboard for an e-commerce company?

- Identification of seasonal buying patterns and high-volume demand periods.
- Tracking return rates and discount margins across product categories to prevent revenue leakage.
- Mapping customer acquisition trends across geographic regions to optimize targeted marketing spend.

10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

Ans- Interactive dashboards replace static interpretations with a shared, exploratory tool. When stakeholders ask "what-if" questions or want deeper detail, analysts can adjust filters and parameters in real time during meetings to uncover answers together, eliminating back-and-forth reporting delays.